Rigorous Derivations in the Octonionic Flavor Framework: Sterile Neutrinos, QFT, and Gravity

This document provides a rigorous synthesis of derivations extending the exceptional-geometric flavor framework. It details how the underlying octonionic algebra and the Quaternionic Kernel Theorem (QKT) naturally give rise to sterile neutrino properties, novel implications for Quantum Field Theory (QFT), and potential extensions into quantum gravity.

1. Derivation of Sterile Neutrino Properties

The framework naturally accommodates sterile neutrinos by identifying them with fundamental elements within the octonionic structure, specifically within the expanded neutral sector of E_6-inspired models.

1.1. Octonionic Embedding and the QKT

In a two-family benchmark of the E_6 model, the neutral fermion basis is:

$$\Psi_N = (\nu_1, \nu_2, \nu_1', \nu_2', \nu_1^c, \nu_2^c, s_1, s_2, \nu_1'^c, \nu_2'^c)^T$$

The right-handed neutrinos (ν^c) and E_6 singlets (s) are the sterile candidates. The Quaternionic Kernel Theorem (QKT) states that a primitive idempotent vacuum $s = \frac{1}{2}(1 + iu)$ in $\mathbb{C} \otimes \mathbb{O}$ uniquely selects a quaternionic subalgebra $\mathbb{H}_L$ [1]. Sterile states are identified as components orthogonal to the active flavor mixing plane or directly associated with the vacuum structure itself [2].

1.2. Mass Generation and the Seesaw Mechanism

The large Majorana masses for sterile states arise from Yukawa couplings involving the octonionic basis elements and the vacuum idempotent, proportional to the high-scale VEVs (e.g., f_1, f_3) that break the E_6 symmetry:

$$\mathcal{L}_{Majorana} = \frac{1}{2} M_{ij} \bar{\Psi}_i^c \Psi_j + h.c.$$

The QKT ensures this large mass hierarchy ($M_R \sim 10^{11} - 10^{16}$ GeV). The light neutrino mass matrix m_ν is then obtained via the seesaw mechanism:

$$m_\nu \approx -m_D M_R^{-1} m_D^T$$

The sterile admixture in light neutrinos is proportional to $m_D M_R^{-1}$. Given the electroweak scale of m_D (~ 100 GeV), the sterile fraction is:

$$P_{sterile} \approx |m_D M_R^{-1}|^2 \sim 10^{-18} - 10^{-28}$$

This rigorously derives the vanishingly small sterile fractions ($\sim 10^{-20}$) observed in phenomenological reconstructions [3].

2. Derivation of QFT Implications from Non-Associativity

The non-associative nature of octonions, where $(a \cdot b) \cdot c \neq a \cdot (b \cdot c)$, necessitates a Non-Associative Quantum Field Theory (NAQFT) [4].

2.1. The Associator and Structure Constants

The degree of non-associativity is quantified by the associator:

$$[a, b, c] = (a \cdot b) \cdot c - a \cdot (b \cdot c)$$

This is expressed via the octonion structure constants f_{abc}. In NAQFT, field operators $\Phi(x)$ take values in the octonion algebra. To define consistent products, specific orderings must be established, modifying standard QFT propagators and vertices.

2.2. Polarization Patterns and Observables

In a perturbative expansion of NAQFT, the structure constants f_{abc} induce unique polarization correlation patterns [5]. Scattering amplitudes depend on combinations of f_{abc}, leading to:

- **Anomalous Couplings:** Deviations from Standard Model predictions.
- **New Observables:** Correlations sensitive to the underlying non-associative algebra.

  Furthermore, the framework suggests pre-quantum, pre-spacetime theories where QFT emerges from an eight-dimensional octonionic space, implying non-associativity is a signature of pre-geometric structure [6].

3. Derivation of Gravitational Extensions

The integration of octonions into gravity offers pathways to quantum gravity and emergent spacetime.

3.1. Octonionic Spacetime and Instantons

Extending spacetime coordinates to octonionic values leads to a non-commutative and non-associative geometry [7]. This modifies dispersion relations and alters causality at the Planck scale. Furthermore, solving self-duality equations for the spin-connection in eight dimensions yields **octonionic gravitational instantons** [8], providing insights into the topological properties of quantum spacetime.

3.2. Supergravity and AdS_3/Qubit Duality

Octonions are manifest in exceptional supergravity theories (e.g., those involving E_6 and E_8) [9]. Recent derivations explore the **AdS_3/Qubit duality**, where the isometry group of AdS_3 is related to the complexified octonion algebra, linking fundamental quantum information (qubits) to spacetime structure [10].

3.3. Emergent Gravity

In the $E_8 \times \omega E_8$ unification model, gravity is not fundamental but an **emergent phenomenon** [11]. The dynamics on the exceptional Jordan algebra $J_3(\mathbb{O}_{\mathbb{C}})$, governed by octonionic triality, give rise to the spacetime metric and gravitational interactions as collective, emergent properties of the underlying algebraic structure.

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